Computational Physics Lectures

Week 04: Molecular dynamics

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Week 4

Molecular dynamics

4.1 Integration

4.1.1 Euler integration

In the Euler method, we update the position and velocity using

$$x(t + \Delta t) = x(t) + v(t)\Delta t,$$
$$v(t + \Delta t) = v(t) + a(t)\Delta t,$$

where $a(t)$ is the acceleration,

$$a(t) = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = \frac{1}{m} F_{\mathrm{res}}(x(t)).$$

Taylor Expansion for Position

The Taylor expansion of $x(t + \Delta t)$ around t is

$$x(t + \Delta t) = x(t) + v(t)\Delta t + \frac{1}{2} a(t)\Delta t^2 + O(\Delta t^3).$$

The Euler update is

$$x_{\mathrm{Euler}}(t + \Delta t) = x(t) + v(t)\Delta t.$$

It neglects terms beginning with $\frac{1}{2} a(t)\Delta t^2$, so its local truncation error is

$$\mathrm{Error}_{\mathrm{Euler}} = \frac{1}{2} a(t)\Delta t^2 + O(\Delta t^3).$$

Thus, if a single Euler step starts from the exact values $x(t)$ and $v(t)$, the error introduced during that step is $O(\Delta t^2)$. This is called the *local truncation error*.

To integrate over a fixed time interval of length T, approximately

$$N = \frac{T}{\Delta t}$$

time steps are required. Each step introduces an error of order Δt^2, and these stepwise errors are propagated through the subsequent steps. Under the usual assumptions that the solution is sufficiently smooth and that the numerical scheme is stable, their combined magnitude therefore scales as

$$N\,O(\Delta t^2) = \frac{T}{\Delta t}\,O(\Delta t^2) = O(\Delta t).$$

The error after integration over a fixed time interval is called the *global error*. Since the global error is proportional to the first power of Δt, the Euler method is said to be *first-order accurate*. Thus, halving Δt should approximately halve the global error when Δt is sufficiently small.

4.1.2 Velocity-Verlet

The velocity-Verlet method is implemented as follows:

1. Position update:
$$x(t + \Delta t) = x(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2.$$

2. Evaluate the new force and acceleration:
$$a(t + \Delta t) = \frac{1}{m}F_{\text{res}}(x(t + \Delta t)).$$

3. Velocity update:
$$v(t + \Delta t) = v(t) + \frac{1}{2}\left(a(t) + a(t + \Delta t)\right)\Delta t.$$

Taylor Expansion for Position

The exact Taylor expansion of the position around t is

$$x(t + \Delta t) = x(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 + \frac{1}{6}\frac{\mathrm{d}a}{\mathrm{d}t}\Delta t^3 + O(\Delta t^4).$$

The velocity-Verlet position update retains all terms in the Taylor expansion through order Δt^2:

$$x_{\text{Verlet}}(t + \Delta t) = x(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2.$$

Thus, the first term omitted by the numerical update is proportional to Δt^3. Therefore, if the step starts from the exact position and velocity, the position after one velocity-Verlet step differs from the exact position by

$$x(t + \Delta t) - x_{\text{Verlet}}(t + \Delta t) = \frac{1}{6}\frac{\mathrm{d}a}{\mathrm{d}t}\Delta t^3 + O(\Delta t^4) = O(\Delta t^3).$$

The local truncation error in the position is consequently $O(\Delta t^3)$. Over a fixed time interval containing approximately $T/\Delta t$ steps, these local errors lead to a global error of order $O(\Delta t^2)$, provided the method is stable. This is why velocity-Verlet is a second-order method.

Taylor Expansion for Velocity

First expand $v(t + \Delta t)$ around t:

$$v(t + \Delta t) = v(t) + a(t)\Delta t + \frac{1}{2}\frac{\mathrm{d}a}{\mathrm{d}t}\Delta t^2 + O(\Delta t^3).$$

In velocity-Verlet, we use the average of the accelerations at t and $t + \Delta t$:

$$v_{\text{Verlet}}(t + \Delta t) = v(t) + \frac{1}{2}\left(a(t) + a(t + \Delta t)\right)\Delta t.$$

Expanding the new acceleration gives

$$a(t + \Delta t) = a(t) + \frac{\mathrm{d}a}{\mathrm{d}t}\Delta t + O(\Delta t^2).$$

Substitution into the velocity update gives

$$\begin{aligned} v_{\text{Verlet}}(t + \Delta t) &= v(t) + \frac{1}{2}\left(a(t) + a(t) + \frac{\mathrm{d}a}{\mathrm{d}t}\Delta t + O(\Delta t^2)\right)\Delta t \\ &= v(t) + a(t)\Delta t + \frac{1}{2}\frac{\mathrm{d}a}{\mathrm{d}t}\Delta t^2 + O(\Delta t^3). \end{aligned}$$

This matches the Taylor expansion of $v(t + \Delta t)$ through second order. The local truncation error is $O(\Delta t^3)$, while the global error over a fixed time interval is $O(\Delta t^2)$. Thus, velocity-Verlet is second-order accurate.

4.1.3 Recap

- **Euler method**: local truncation error $O(\Delta t^2)$ and global error $O(\Delta t)$; first-order accurate.
- **Velocity-Verlet method**: local truncation error $O(\Delta t^3)$ and global error $O(\Delta t^2)$; second-order accurate.

For sufficiently small Δt, velocity-Verlet is generally substantially more accurate than Euler at the same time step and is much better suited to long molecular-dynamics trajectories.

4.2 Thermostat

The method of molecular dynamics is nicely introduced in the paper by Katharina Vollmayr-Lee, and there is little to add. One important issue, however, is the choice of thermostat.

The paper introduces a procedure in which thermal velocities are drawn and then rescaled. It is useful first to describe this procedure and then to explain why, for the small systems considered in this course, we shall use a simpler approach.

For N atoms in three dimensions, the procedure is:

1. For every atom i and spatial direction $\mu \in \{x, y, z\}$, draw $v_{i,\mu}$ from a normal distribution with standard deviation $\sigma_i = 1$.

2. Account for the particle masses by rescaling

$$v_{i,\mu} \to \frac{1}{\sqrt{m_i}} v_{i,\mu}.$$

3. If zero total momentum is required, remove the center-of-mass velocity. For each spatial direction,

$$V_{\mathrm{CM},\mu} = \frac{\sum_i m_i v_{i,\mu}}{\sum_i m_i},$$

and hence

$$v_{i,\mu} \to v_{i,\mu} - V_{\mathrm{CM},\mu}.$$

4. Rescale all velocities to obtain the desired kinetic energy:

$$v_{i,\mu} \to \sqrt{\frac{f k_{\mathrm{B}} T}{\sum_{j,\nu} m_j v_{j,\nu}^2}}\, v_{i,\mu},$$

where f is the number of unconstrained velocity degrees of freedom and the sum in the denominator runs over all particles j and spatial directions ν.

For N particles in d dimensions, $f = dN$ if center-of-mass motion is retained. If center-of-mass motion is removed, d constraints are imposed and

$$f = dN - d = d(N-1).$$

Thus, it is the number f of unconstrained degrees of freedom, rather than automatically $3N$, that must be used in the rescaling factor.

For a small system, exact velocity rescaling is not equivalent to canonical sampling: it fixes the instantaneous kinetic energy at its target value and suppresses the natural kinetic-energy fluctuations. In this course, where we consider only a few degrees of freedom, we shall instead draw each velocity component directly from the Maxwell–Boltzmann distribution,

$$v_{i,\mu} \sim \mathcal{N}\left(0, \frac{k_{\mathrm{B}} T}{m_i}\right),$$

that is, from a normal distribution with standard deviation

$$\sigma_i = \sqrt{\frac{k_{\mathrm{B}} T}{m_i}}.$$

If required, center-of-mass motion can subsequently be removed using the expression above. Drawing the velocities once initializes the system at temperature T; repeated redraws during a trajectory constitute a stochastic thermostat and require a specified redraw frequency.

4.3 Simulated annealing

In molecular dynamics (MD) simulations, simulated annealing is a technique used to search for low-energy conformations of molecular systems by mimicking the physical process of annealing

in materials science. The basic idea is to heat the system to a high temperature, allowing it to overcome energy barriers and explore a wide range of configurations, and then gradually cool it so that it may settle into a low-energy state.

The procedure involves the following steps:

1. **Initialization:** Initialize the system at a high temperature and assign random velocities according to a Maxwell–Boltzmann distribution.
2. **High-temperature dynamics:** Allow the system to evolve at the high temperature using a molecular-dynamics integration method such as velocity-Verlet, together with a specified thermostatting procedure.
3. **Cooling schedule:** Gradually reduce the target temperature according to a predefined schedule, such as exponential or linear decay. The thermostat is applied using the current target temperature.
4. **Low-temperature dynamics:** As the system cools, it explores lower-energy regions of the energy landscape and may settle into a low-energy minimum.
5. **Final state:** Once the system has reached a sufficiently low temperature, stop the simulation and analyze the final configuration. The resulting structure is a candidate low-energy configuration.

Simulated annealing in MD can be applied to a wide range of problems, including protein folding, ligand binding, and optimization of material structures. A finite-time simulated-annealing run does not guarantee that the global minimum is found. Its success depends on the initial temperature, cooling schedule, simulation time, energy landscape, and ability of the system to explore configuration space. It is therefore useful to perform several independent runs with different random initial velocities.